

Total No. of Questions : 10]

SEAT No. :

P3668

[5561]-301

[Total No. of Pages : 2

B.E. (I.T.)

SOFTWARE MODELLING AND DESIGN

(Semester - I) (2012 Course)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
- 2) *Diagrams should be drawn with UML 2.0 notations.*
- 3) *Figures to the right indicate full marks.*

Q1) a) In the context of class diagram show a generalization and composite relationship with the help of one example each. [6]

b) In the context of state diagram, define state, trigger, guard and effect. [4]

OR

Q2) a) In the context of class diagram what is a qualified association and association class. Show using an example. [6]

b) In the context of use case diagram show the include and extend relationship using a diagram. Draw the diagram only with all relevant parts labelled. [4]

Q3) a) With the context of activity diagram, elaborate decision and merge with the help of an example. [6]

b) The use case login accepts the userid and password from the user. The system validates them and displays the message that 'User Id or password is invalid'. Draw a sequence diagram for this. [4]

OR

Q4) a) Draw a state diagram of ordering Pizza using a web based system. Draw the states from placing the order till the delivery. [6]

b) Draw a use case diagram with appropriate relationships and notation for the following description. Consider an online travel planner software. Through this software, the user can book bus tickets, provide payment information, provide address, book a car on rent, book a hotel room. It is mandatory to provide payment information and provide address for booking of bus tickets, car on rent and a hotel room. [4]

P.T.O.

- Q5) a)** Write a note on Making a Reuse Plan from the context of system design. [8]
b) Which are the boundary conditions & how they are handled? [8]

OR

- Q6) a)** Describe one way of breaking a system into subsystems. [8]
b) Describe allocation of subsystems in System Design. [8]

- Q7) a)** Write the classification, motivation, class diagram and uses of adapter design pattern. [8]
b) Write the classification, motivation, class diagram and uses of state design pattern. [8]

OR

- Q8) a)** Write the classification, motivation, class diagram and uses of strategy design pattern. [8]
b) Write the classification, motivation class diagram and uses of observer design pattern. [8]

- Q9) a)** Draw a flow chart of test driven development. [8]
b) Features of printer are to be tested & its specification is as follows. [10]
It prints the document in black & white, colour. It has an on/off button. It accepts A4 paper one at a time for printing. It has two LED lights. Green light shows normal printing operation & red light shows problem with printing. It has two cables. One is power cable & other is data cable, which is connected to the CPU. Write at least five test cases to check that the Printer machine works properly. [10]

OR

- Q10) a)** Differentiate between black box and white box testing. [8]
b) Elaborate the V curve of testing. [10]

